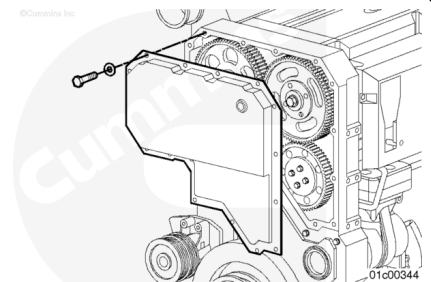


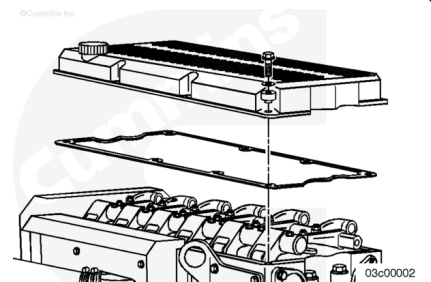
Trainings ▾

Preparatory Steps

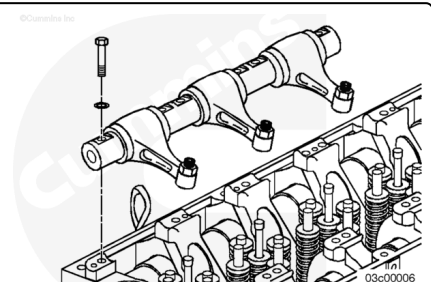
Remove the upper gear cover. Refer to Procedure 001-079 (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>) in Section 1.



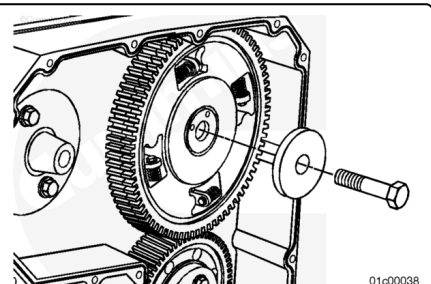
Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>).



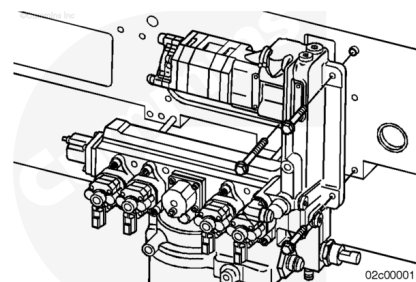
Remove the rocker lever shafts. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html>) in Section 3.



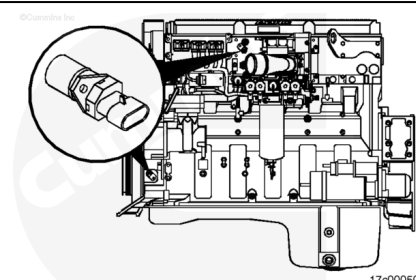
Remove the injector camshaft scissor gear. Refer to Procedure 001-088 (</qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html>) in Section 1.



Remove the integrated fuel system module. Refer to Procedure 005-073 in Section 5
(/qs3/pubsys2/xml/en/procedures/10/10-005-073-tr.html).



Remove the camshaft sensor. Refer to Procedure 019-038
(/qs3/pubsys2/xml/en/procedures/72/72-019-038.html) in
Section 19.

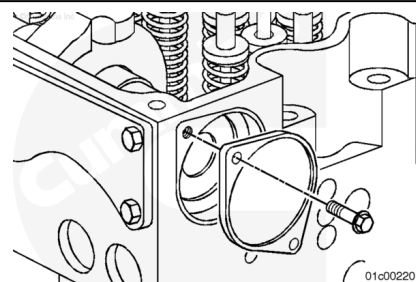


Remove

Remove the injector camshaft cover plate capscrews.

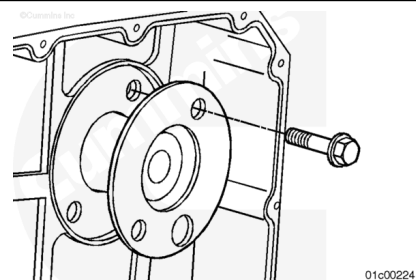
Remove the injector cover plate and gasket.

Discard the gasket.



Remove the two capscrews and the injector camshaft retainer
plate and o-rings.

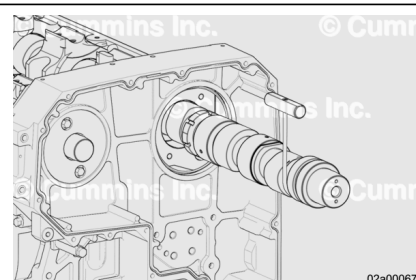
Discard the o-rings.



⚠ WARNING ⚠

**To reduce the possibility of personal injury, wear protective
gloves when handling parts with sharp edges.**

⚠ WARNING ⚠



This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Use caution when removing the camshaft to avoid damaging the camshaft bushings.

Remove the injector camshaft.

Use the Camshaft Installation and Removal Kit, Part Number 5395348, to pull and guide the camshaft out of the cylinder head until three journals remain in the cylinder head. Carefully remove the remainder of the camshaft by hand.

Clean and Inspect for Reuse

⚠ WARNING ⚠

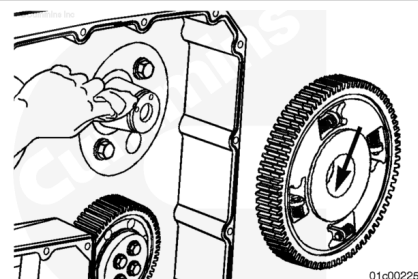
To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

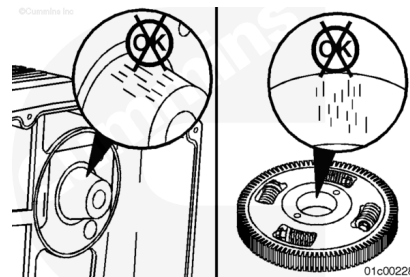


Clean the injector camshaft and the bore of the injector camshaft scissor gear with ENVIROSOL™ 655 solvent, or equivalent, and a lint-free cloth.

Clean all debris from the cylinder head and the camshaft bushings.

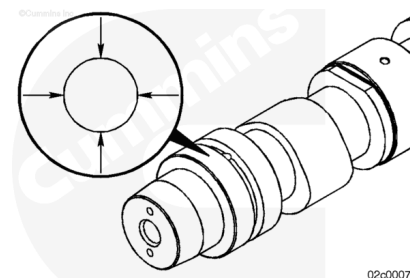
Check the injector camshaft and injector camshaft scissor gear for fretting damage.

The camshaft and gear **must** be replaced if damaged.



Measure the seven injector camshaft bearing journals.

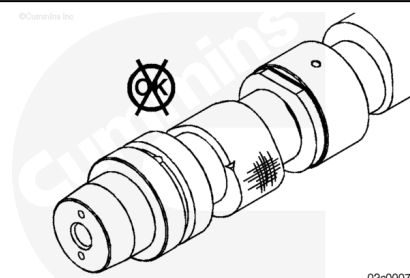
Injector Camshaft Bearing Journal Outside Diameter		
mm		in
84.987	MIN	3.3459
85.013	MAX	3.3469



If the camshaft bearing journals are **not** within specifications, the camshaft **must** be replaced.

Inspect the camshaft lobes for cracks, scratches, or other damage.

Refer to the Camshaft Reuse Guidelines for Cummins® Engines with Roller Followers or Roller Tappets, Service Bulletin 3666052 (</qs3/pubsys2/xml/en/bulletin/3666052.html>), for further reuse guidelines.



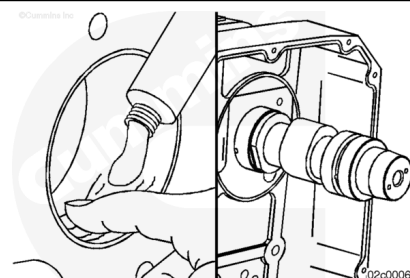
Install

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Use caution when installing the camshaft to avoid damaging the camshaft bushings.

Coat the inside of the injector camshaft bearings with Lubriplate™ prior to installing camshaft.

Install the injector camshaft.

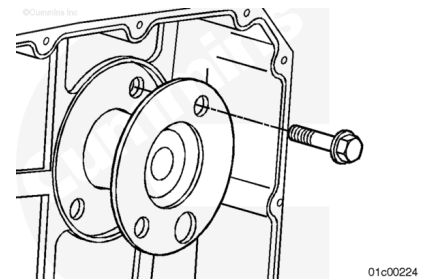
Install the camshaft by hand until three journals are in the cylinder head. Use the Camshaft Installation and Removal Kit, Part Number 5395348, to pull and guide the camshaft into the cylinder head.

Install new o-rings on the retainer plate.

Install the retainer plate and capscrews.

Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]

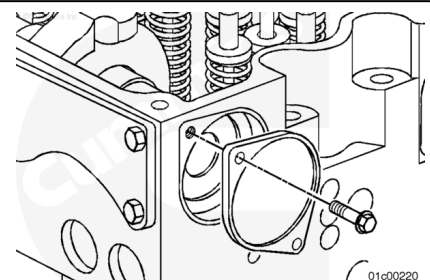


Install the injector camshaft cover plate with a new gasket.

Install the capscrews.

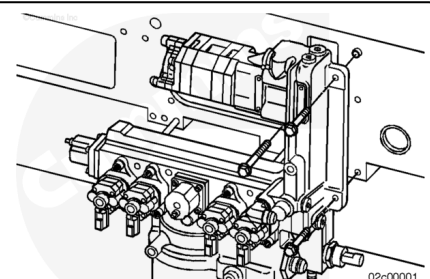
Tighten the capscrews.

Torque Value: 40 n•m [30 ft-lb]

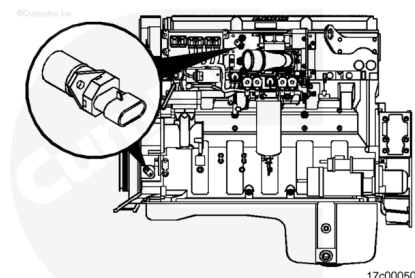


Finishing Steps

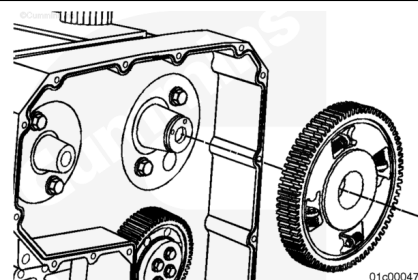
Install the integrated fuel system module. Refer to Procedure 005-073 (/qs3/pubsys2/xml/en/procedures/10/10-005-073-tr.html) in Section 5.



Install the camshaft sensor. Refer to Procedure 019-038 in Section 19. (</qs3/pubsys2/xml/en/procedures/72/72-019-038.html>)

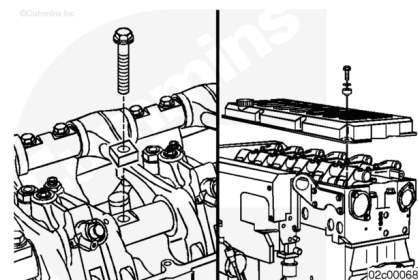


Install the injector camshaft scissor gear. Refer to Procedure 001-088 (</qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html>) in Section 1.

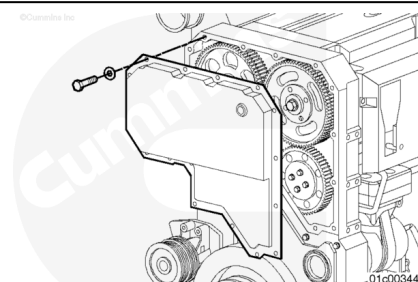


Install the rocker lever shafts. Refer to Procedure 003-009 (</qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html>) in Section 3.

Install the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html>) in Section 3.



Install the upper gear cover. Refer to Procedure 001-079 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html>).



Operate the engine to normal operating temperature. Check for leaks.

